Portfolio overview

Clients	Observed Default Rate	Exposure (NT$)	Mean Model PD
9 000	22,1%	1 507 047 680	21,9%

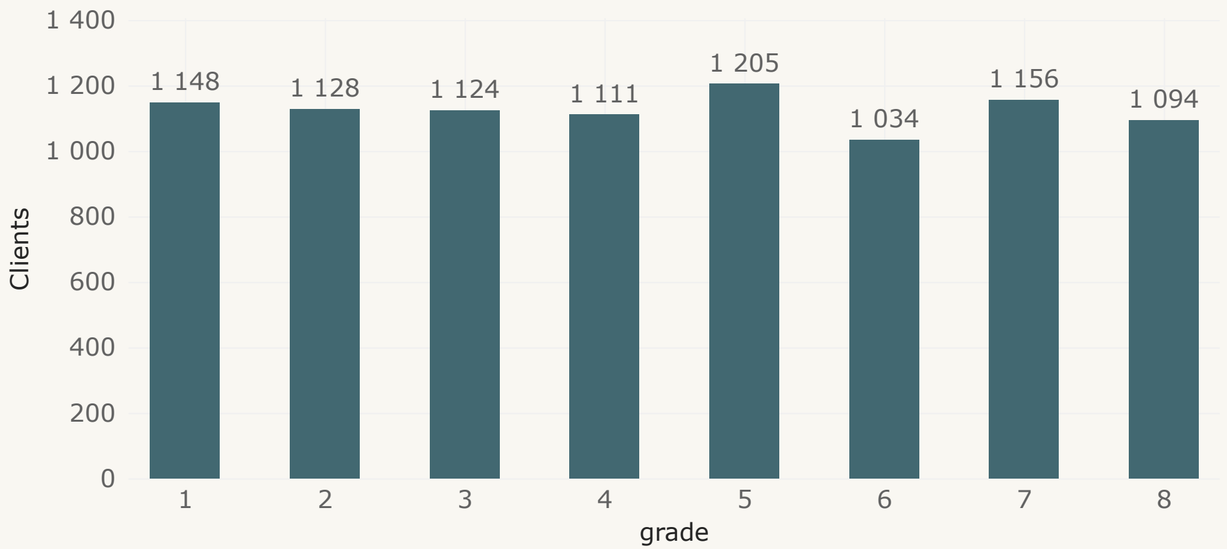
segment_id

- [ ] Graduate school
- [ ] High school
- [ ] Other
- [ ] University

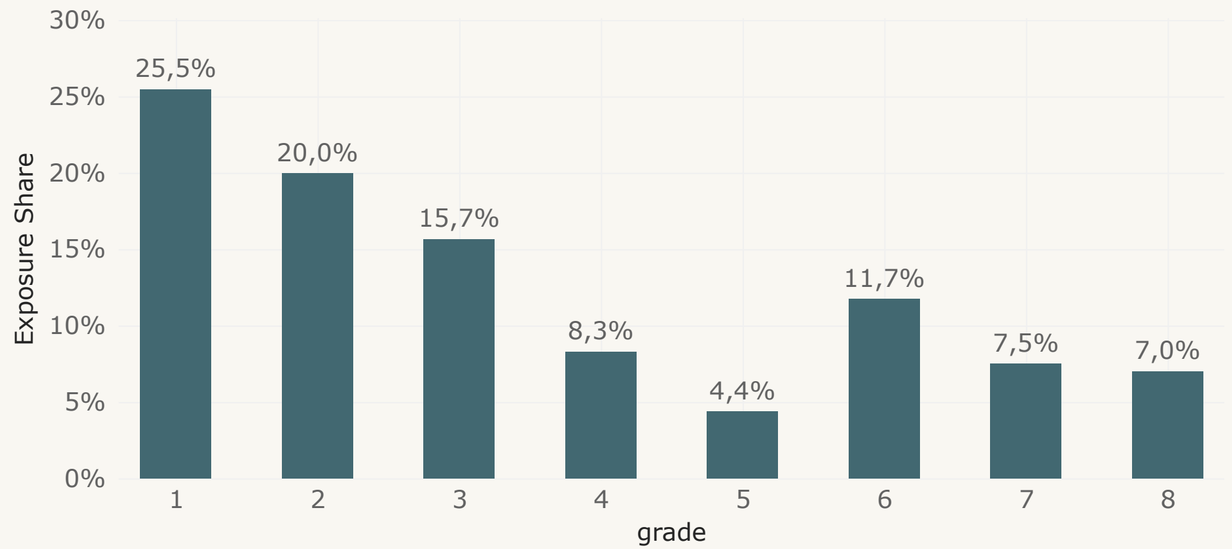
age_band

- [ ] under 30
- [ ] 30-39
- [ ] 40-49
- [ ] 50 plus

Clients by grade



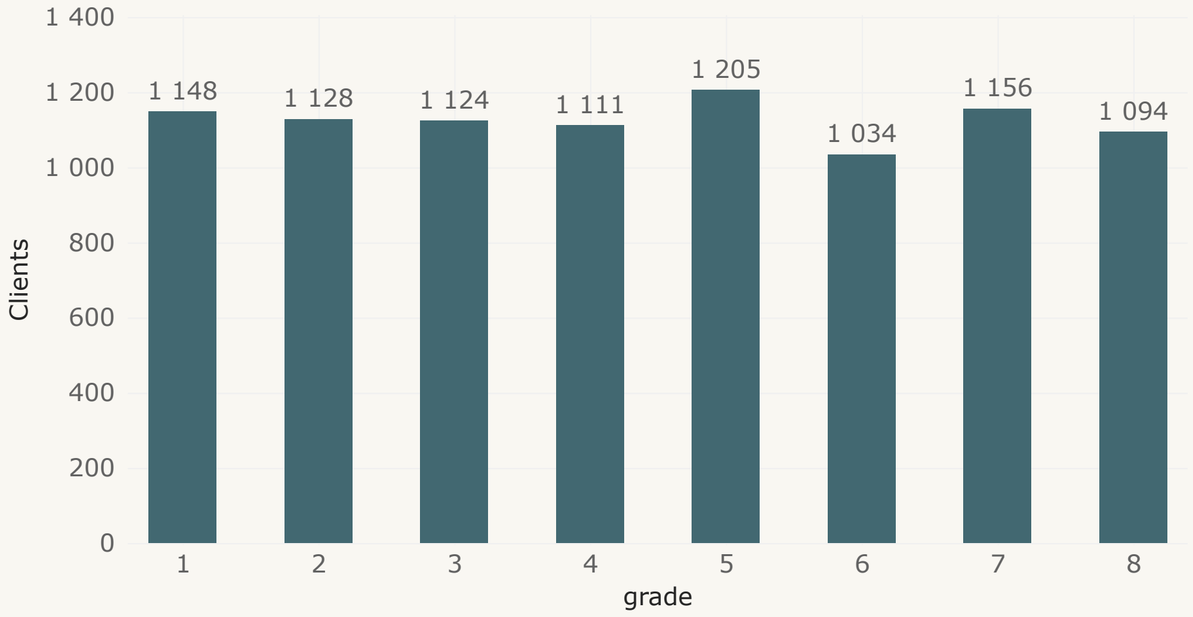
Exposure share by grade



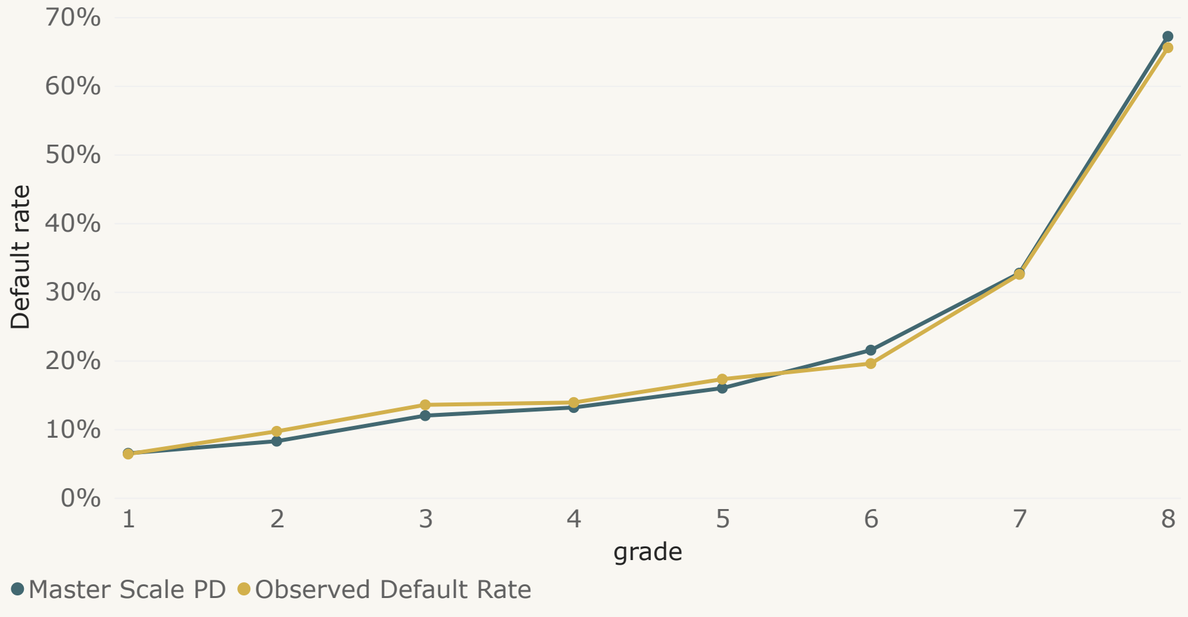
Test sample, 9,000 clients, one-month PD, UCI credit card data

Calibration

Clients by grade



Master scale PD vs observed default rate



grade	clients	defaults	observed_rate	master_scale_pd	binomial_p
8	1094	717	65,5%	67,2%	0,247
7	1156	376	32,5%	32,7%	0,925
6	1034	202	19,5%	21,5%	0,130
5	1205	208	17,3%	16,0%	0,223
4	1111	154	13,9%	13,1%	0,477
3	1124	152	13,5%	12,0%	0,107
2	1128	109	9,7%	8,2%	0,083
1	1148	73	6,4%	6,5%	0,952

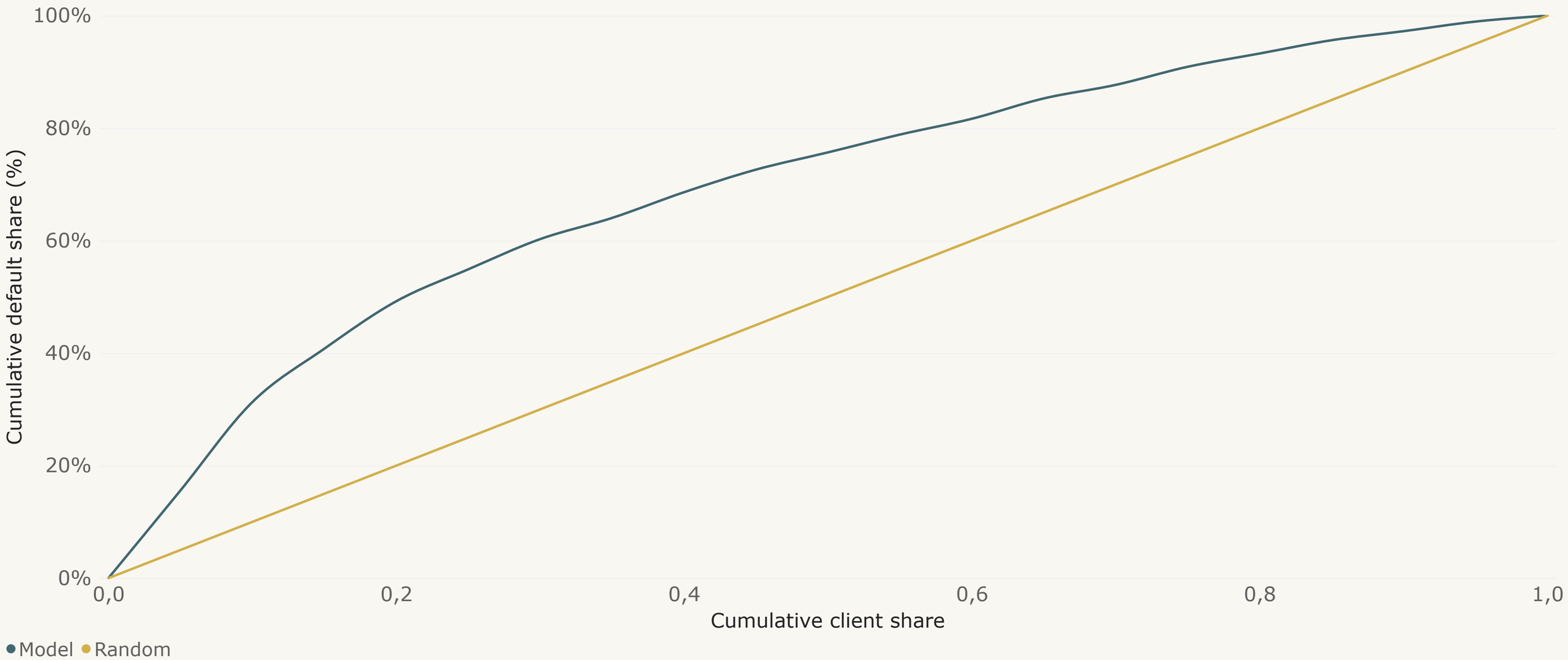
Defaults vs Expected
1,008

No grade is rejected by the exact binomial test at 5% (no correction for eight tests).

Test sample, 9,000 clients, one-month PD, UCI credit card data

Discrimination

Cumulative Accuracy Profile (CAP): Model vs Random



Gini 0.505 on the test sample (Python and SQL agree). The 20% riskiest clients by PD hold 49% of the defaults.

Test sample, 9,000 clients, one-month PD, UCI credit card data

Benchmark

age_band	under 30		30-39		40-49		50 plus	
segment_id	**Rate**	**Gap**	**Rate**	**Gap**	**Rate**	**Gap**	**Rate**	**Gap**
High school	26,8%	4,3%	23,2%	3,3%	25,1%	1,1%	26,8%	0,8%
University	25,0%	2,5%	21,6%	1,8%	25,3%	1,4%	29,0%	3,1%
Graduate school	18,5%	-3,9%	17,3%	-2,5%	21,6%	-2,3%	21,2%	-4,7%
Other	3,6%	-18,9%	5,1%	-14,7%	7,5%	-16,4%	15,4%	-10,6%

Gap = segment default rate minus the rate of all segments in the same age band. Other: only 120 clients.

Test sample, 9,000 clients, one-month PD, UCI credit card data